ECE 329
Lecture 17

Magnetic Materials

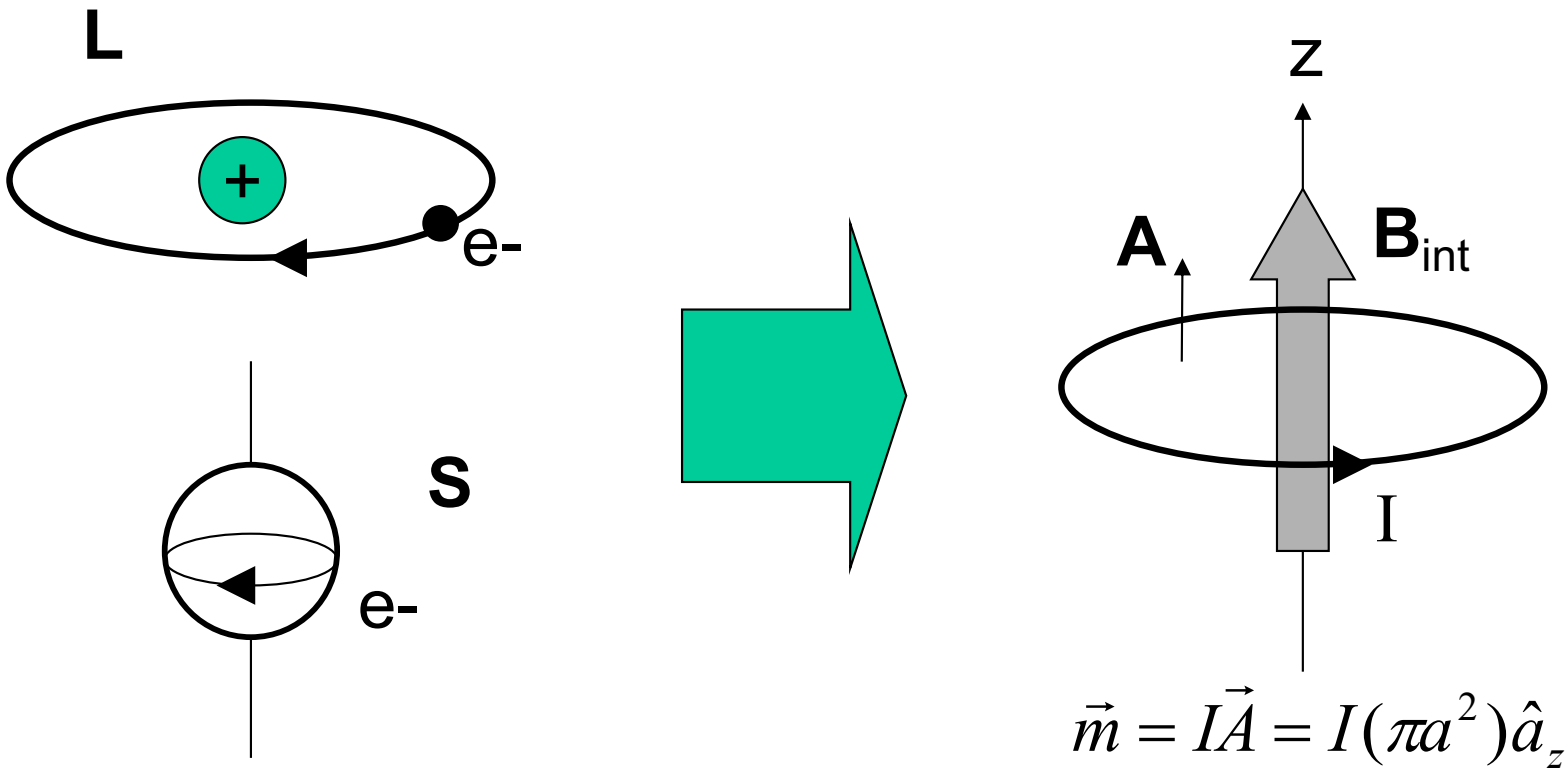
Maxwell's equations in material media



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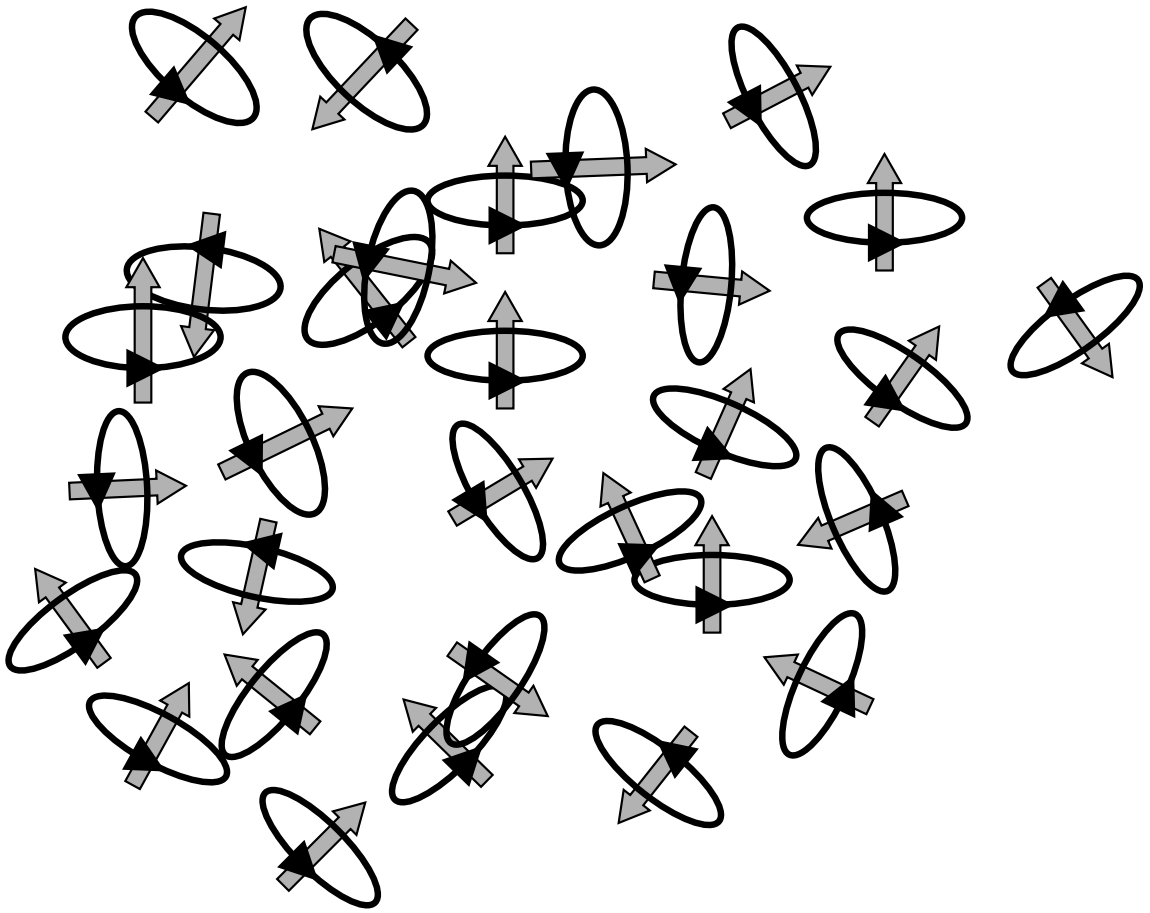
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Magnetic Moments at the atomic scale



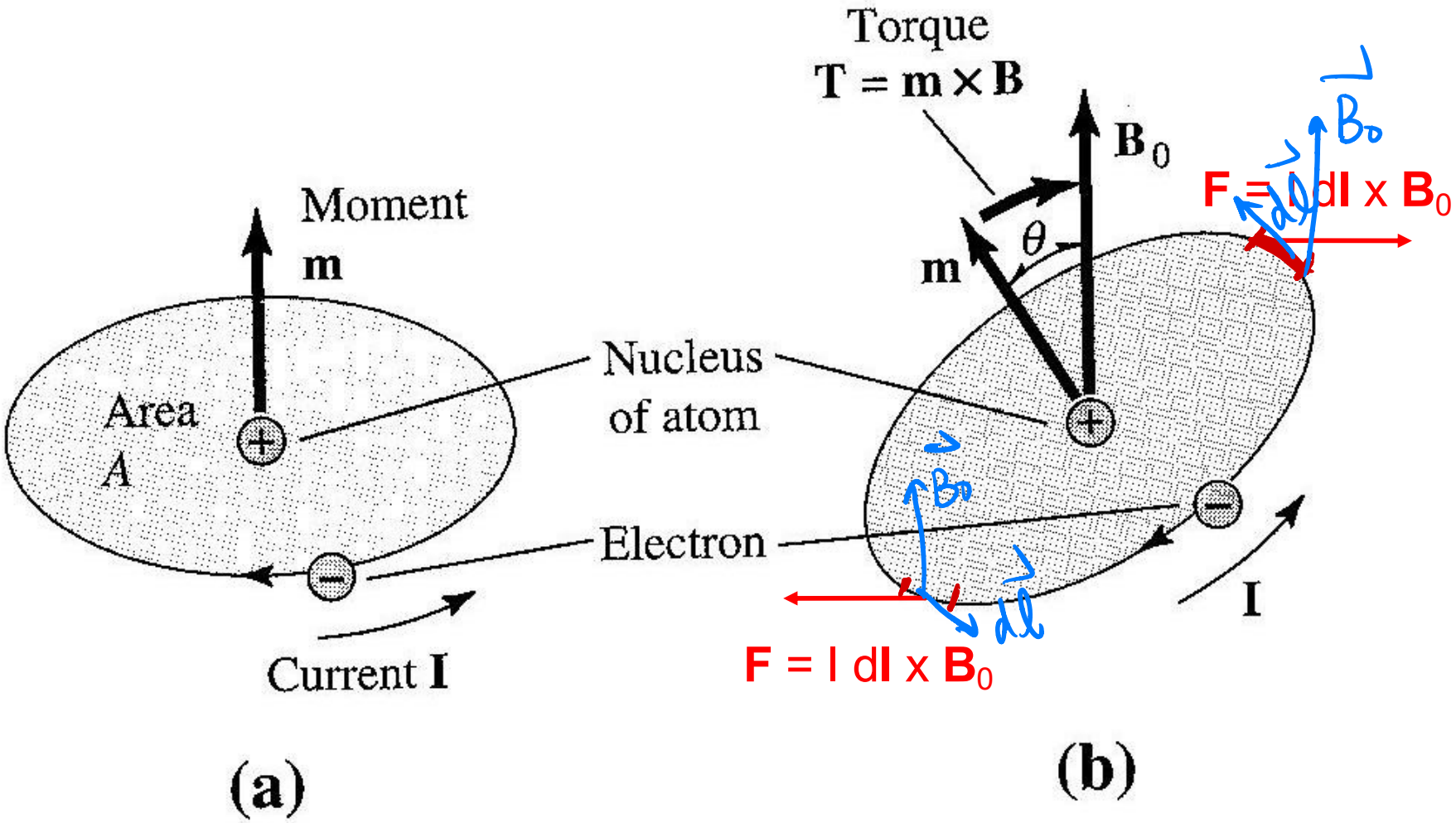
Internal magnetic fields are produced by electrons orbiting the nucleus, **L**, or by the internal spin of electrons, **S**
The atom has a magnetic dipole moment, **m**

Net magnetic moment

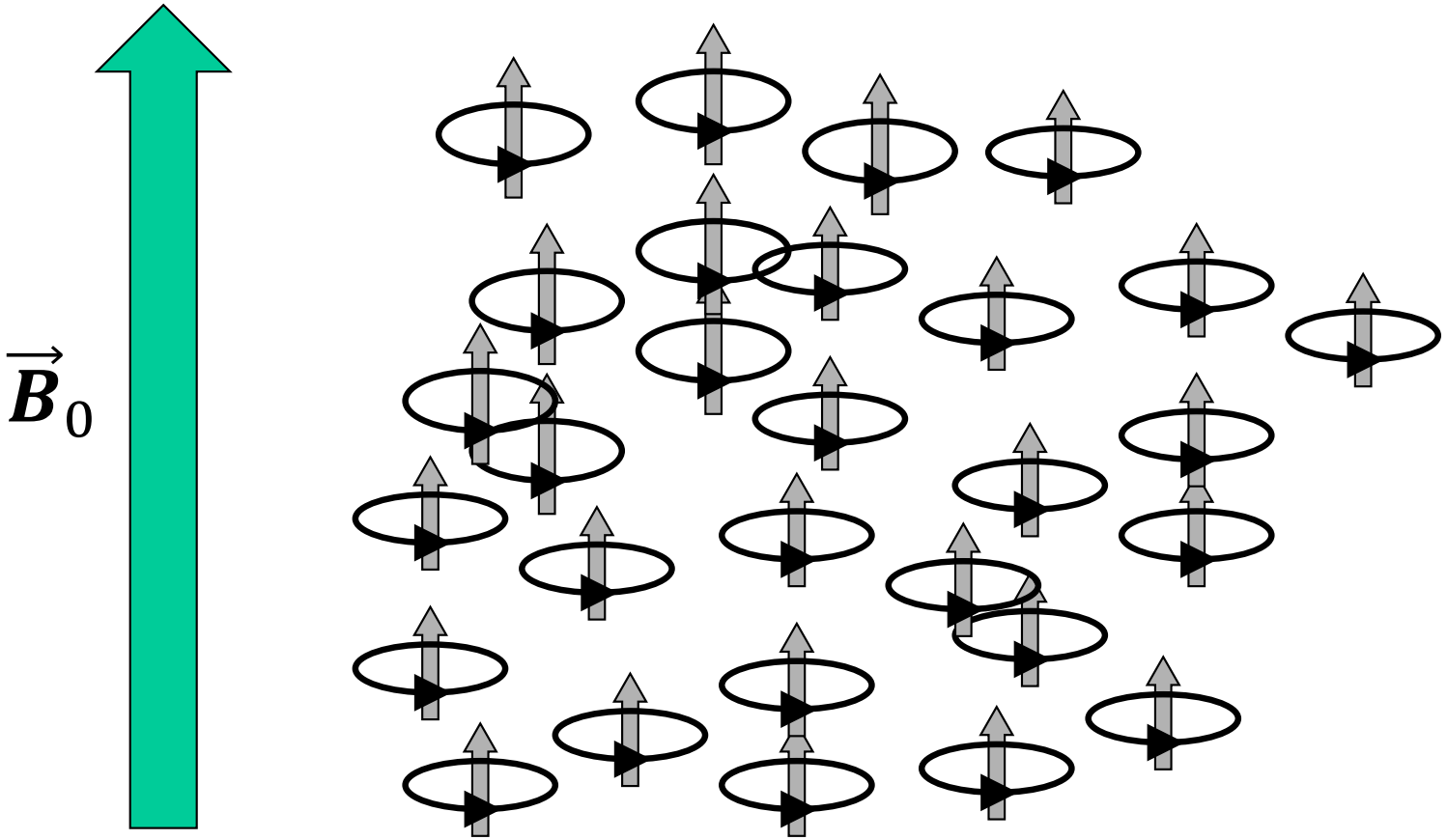


A volume of material contains many magnetic moments. They might be randomly oriented.

External **B**-field can rotate Magnetic Moments or change Orbital Velocity



Net magnetic moment



After momentarily applying an external field, they can get magnetized and are mostly aligned in one direction.
The internal field can keep them aligned to each other

Magnetization Vector

$$\vec{M} = N\vec{m}$$

Magnetization or magnetic dipole moment per unit volume

N = # atoms per unit volume

m = average dipole moment per molecule

Units for **M**: $(1/m^3)\ (A*m^2) = A/m$ → same $\vec{H}$

$$\vec{B}_{\text{int}} = \mu_0 \vec{M}$$

is the magnetic flux per unit area from the dipoles

Magnetic Susceptibility

$$\vec{B}_{total} = \vec{B}_0 + \vec{B}_{int}$$

The INTERNAL magnetic flux is INDUCED by the application of an EXTERNAL magnetic field

Total flux = Applied + Secondary

$$\vec{B}_{total} = \mu_0(\vec{H}_{ext} + \vec{M})$$

Magnetization: $\vec{M} = \chi_m \vec{H}_{ext}$

$\vec{P} = \varepsilon_0 \chi_e \vec{E}$

$[C/m^2]$ $[V/m]$

Definition of magnetic susceptibility (has no units)

Relative Magnetic Permeability

Magnetic

$$\vec{B}_{total} = \mu_0(\vec{H}_{ext} + \vec{M})$$

$$\vec{M} = \chi_m \vec{H}_{ext}$$

no μ_0

$$\vec{B}_{total} = \mu_0(1+\chi_m)\vec{H}_{ext}$$

$$\vec{B} = \mu\vec{H}$$

$$\mu = \mu_0(1+\chi_m) = \mu_0\mu_r$$

$$\mu_r = 1+\chi_m$$

dielectric

$$\vec{D} = \varepsilon_0\vec{E}_{tot} + \vec{P}$$

$$\vec{P} = \varepsilon_0\chi_e\vec{E}_{tot}$$

$$\vec{D} = \varepsilon_0(1+\chi_e)\vec{E}_{tot}$$

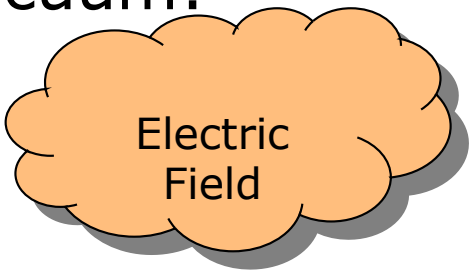
$$\vec{D} = \varepsilon\vec{E}_{tot}$$

$$\varepsilon = \varepsilon_0(1+\chi_e) = \varepsilon_0\varepsilon_r$$

$$\varepsilon_r = 1+\chi_e$$

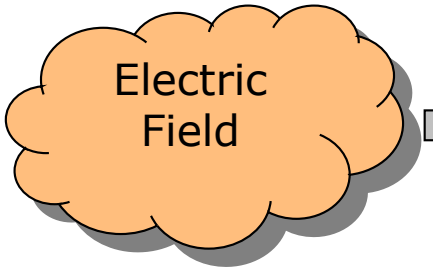
Recall: Displacement field $\vec{D}$ and Polarization $\vec{P}$

In vacuum:



Displacement
(electric flux density)
$\vec{D} = \epsilon_o \vec{\mathbf{E}}$

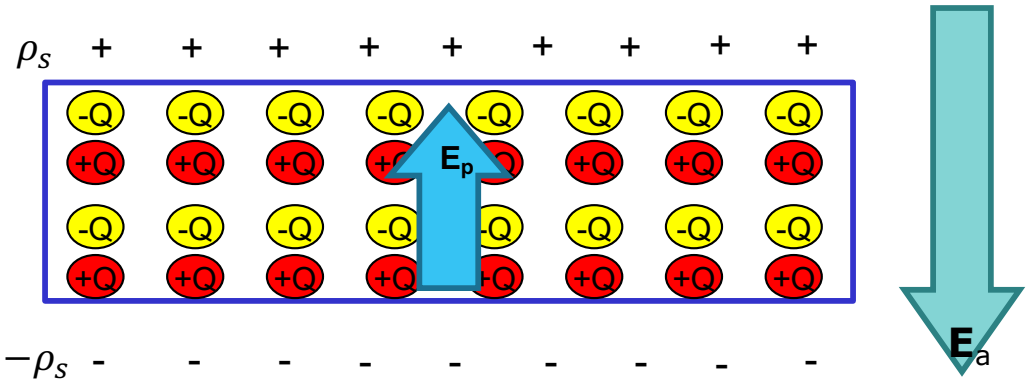
In dielectric:



Displacement
(electric flux density)
$\vec{D} = \vec{P} + \epsilon_o \vec{\mathbf{E}}$

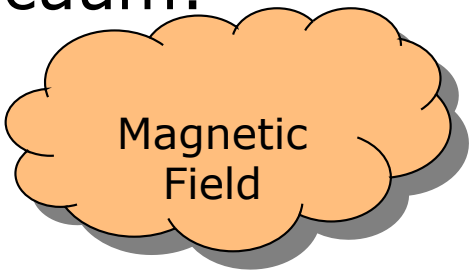
Item $\epsilon_o \vec{\mathbf{E}}$: is the same as the case in vacuum

Item "$\vec{P} = \epsilon_0 \chi_e \vec{E}_{tot}$": is the polarization due to contribution of dipole formation in dielectric under applied $\vec{\mathbf{E}}$ field



Magnetic Flux Density $\vec{B}$ and Magnetic Polarization $\vec{M}$

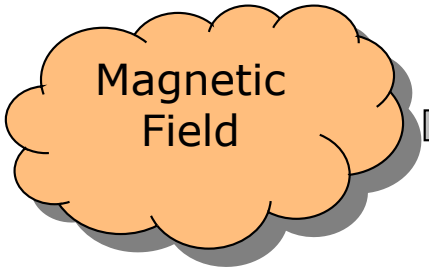
In vacuum:



Magnetic flux density

$$\vec{B} = \mu_o \vec{H}$$

In magnetic material:

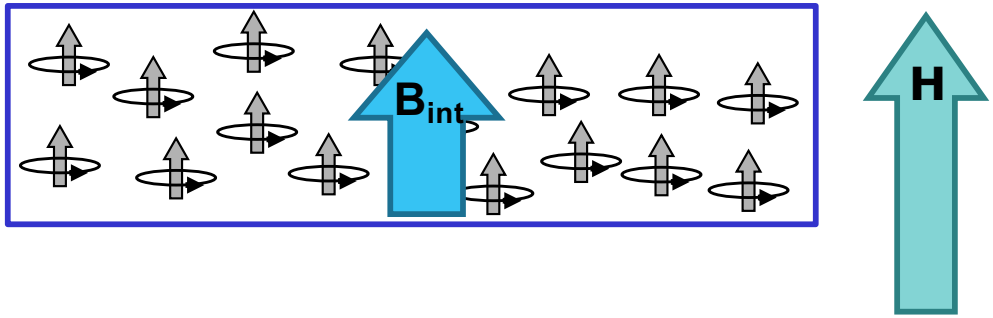


Magnetic flux density

$$\vec{B} = \mu_o \vec{M} + \mu_o \vec{H}$$

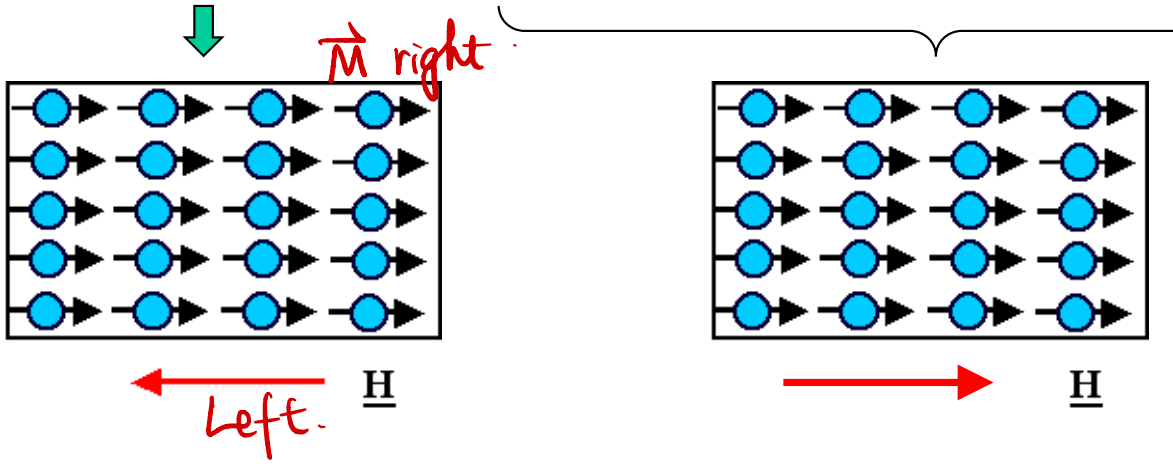
Item $\mu_o \vec{H}$: is the same as the case in vacuum

Item "$\vec{M} = \chi_m \vec{H}_a$": is the magnetization vector inside magnetic material under applied $\vec{H}$ field



Comparison of Dia, Para and Ferro Magnetic materials

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Magnetization	Opposes external field	Weakly aligns with field	Strongly aligns with field
Susceptibility	Negative and small	Positive and small	Positive and large
Retains Magnetism?	No	No	Yes (can be permanent)
Examples	Copper, Bismuth	Aluminum, Platinum	Iron, Nickel, Cobalt



Diamagnetic Materials (χ_m<0)

- Diamagnetic materials repel magnetic lines of force.



- Magnetic susceptibility is a NEGATIVE number

$$\chi_m < 0$$

Examples:

Copper	χ_m = -0.94x10^{-5}
Lead	χ_m = -1.70x10^{-5}
Water	χ_m = -0.88x10^{-5}

Paramagnetic Materials ($\chi_m > 0$)
Positive susceptibility

- Paramagnetic materials attract magnetic lines of force.
- With $\mathbf{H}_{ext}$, domains reorient themselves to generate $\mathbf{B}_{int}$ that ALIGNS WITH H_{ext}



- Magnetic susceptibility is a POSITIVE number

$$\chi_m > 0$$

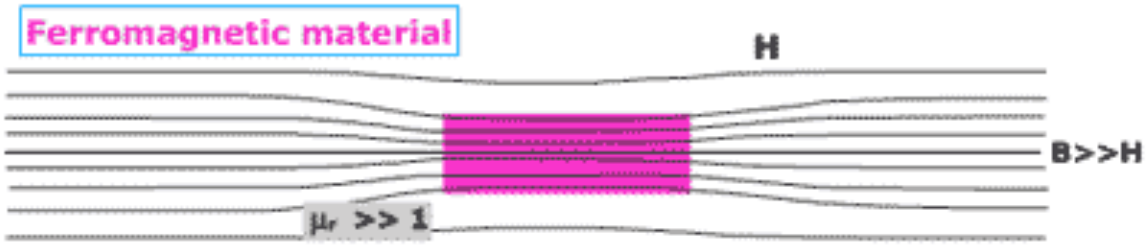
Examples:

Platinum	$\chi_m = +2.90 \times 10^{-5}$
Aluminum	$\chi_m = +2.10 \times 10^{-5}$
Liquid Oxygen	$\chi_m = +3.50 \times 10^{-5}$

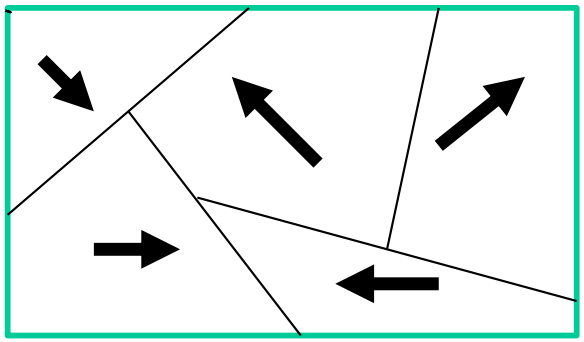
Ferromagnetic Materials ($\chi_m >> 1$)

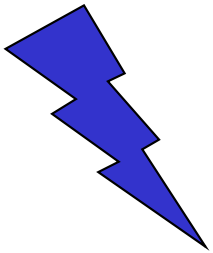
- Microscopic "domains" that have strongly oriented magnetic dipoles
- Under an externally applied H field, the domains can orient coherently (i.e. in the same direction)
- Using a strong enough applied field, the domain orientation can become permanent
- Very high susceptibility

$\mu_r - 1 = \chi_m >> 1$

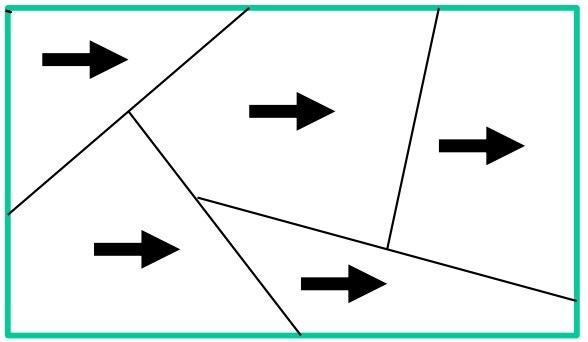


Example: Iron, nickel, cobalt and their alloys.





Apply External Magnetic Field



Material		$\mu_r = 1 + \chi_m$
Bismuth	Diamagnetic	0.99983
Silver	Diamagnetic	0.99993
Lead	Diamagnetic	0.99993
Copper	Diamagnetic	0.999991
Water	Diamagnetic	0.999991
Vacuum	Nonmagnetic	1†
Air	Paramagnetic	1.000000
Aluminum	Paramagnetic	1.00002
Palladium	Paramagnetic	1.0008
2-81 Permalloy powder (2 Mo, 81 Ni)‡	Ferromagnetic	130
Cobalt	Ferromagnetic	250
Nickel	Ferromagnetic	600
Ferroxcube 3 (Mn-An-ferrite powder)	Ferromagnetic	1,500
Mild steel (0.2 C)	Ferromagnetic	2,000
Iron (0.2 impurity)	Ferromagnetic	5,000
Silicon Iron (4 Si)	Ferromagnetic	7,000
78 Permalloy (78.5 Ni)	Ferromagnetic	100,000
Mumetal (75 Ni, 5 Cu, 2 Cr)	Ferromagnetic	100,000
Purified iron (0.05 impurity)	Ferromagnetic	200,000
Superalloy (5 Mo, 79 Ni)§	Ferromagnetic	1,000,000

$\chi_m < 0$.

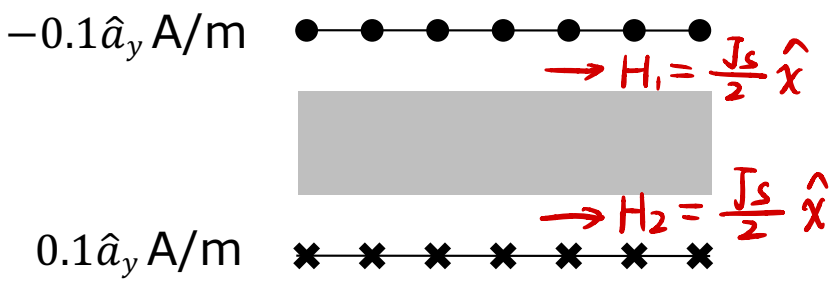
$\chi_m > 0$ small

$\chi_m \gg 1$.

†By definition.
‡Percentage composition. Remainder is iron and impurities.
§Used in transformer applications with continuous tape-wound (gapless) cores.

Example: Magnetization

- An infinite plane ferromagnetic slab ($\mu_r = 100$) lies between two infinite plane sheets of uniform current density of $\vec{J} = -0.1\hat{a}_y$ A/m. Find **H**, **B**, and **M** inside the slab and compare to if the slab were non-magnetic ($\mu_r = 1$).



$\vec{H} = \vec{H}_1 + \vec{H}_2 = J_s\hat{x} = \vec{H}_{slab}$

$\hat{n}\times(\vec{H}_1 - \vec{H}_2) = J_s$

no J_s on slab.

$\vec{B}_{slab} = \mu_0\mu_r\vec{H}_{slab} = 100\mu_0 J_s\hat{x}$

$\vec{M} = \chi_m\vec{H} = 99 J_s\hat{x}$

$\vec{B} = \mu_0\vec{M} + \mu_0\vec{H}$

$\chi_m = \mu_r - 1 = 99$

Hint:

$$\vec{H} = \left(\frac{\vec{J}}{2}\right)\hat{x}\times\hat{a}_n = -\left(\frac{J}{2}\right)\hat{a}_x$$



$\mu_r = 1$

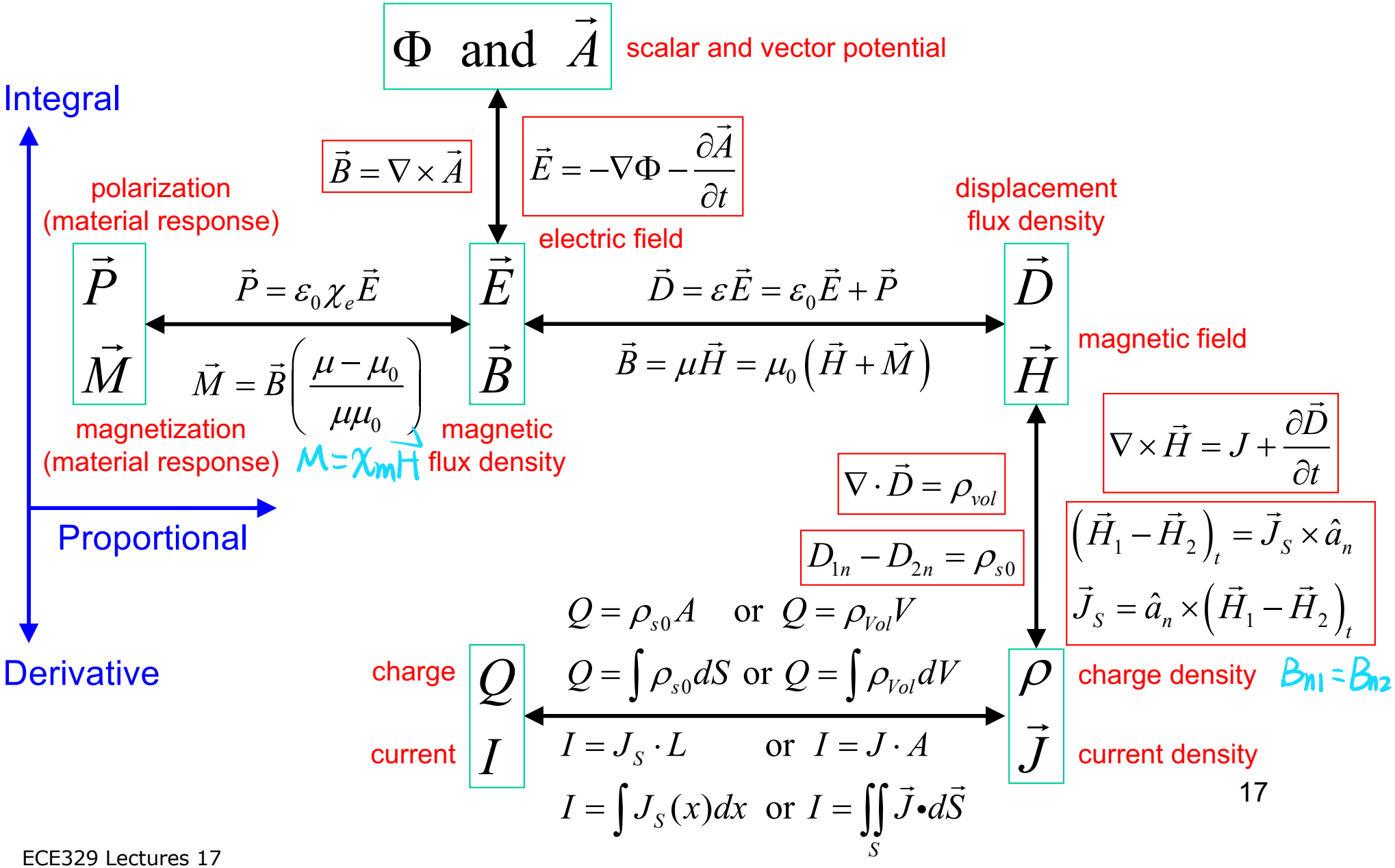
$\vec{H} = J_s\hat{x}$ same.

$\vec{B} = \mu_0\vec{H} = \mu_0 J_s\hat{x}$

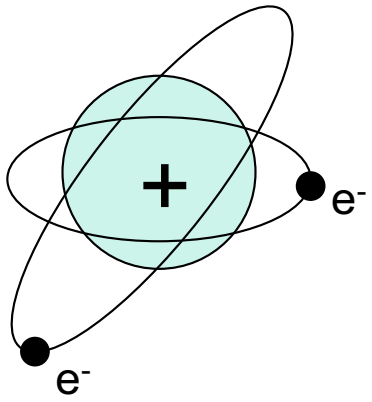
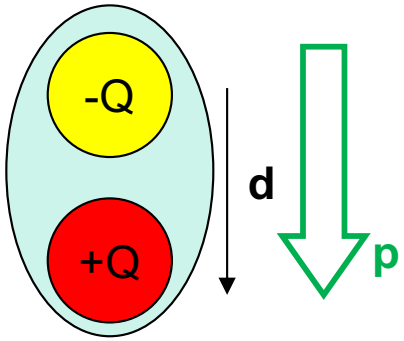
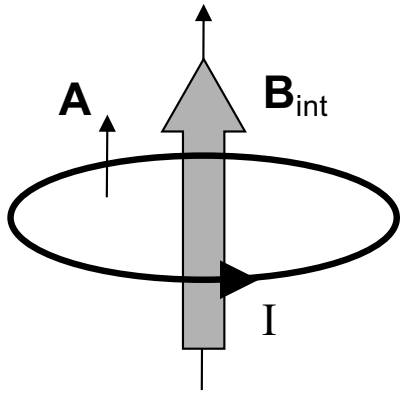
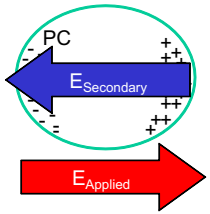
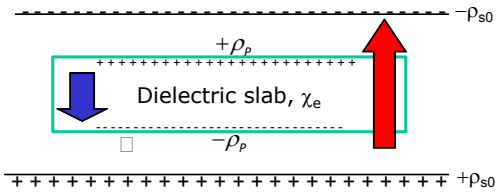
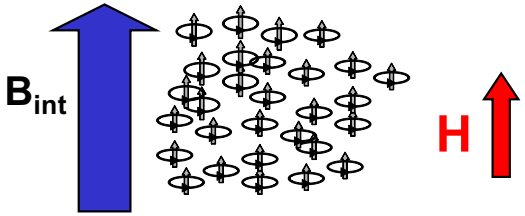
$\vec{M} = 0$ $\chi_m = 0$

$$\vec{H} = \left(\frac{\vec{J}}{2}\right)\hat{x}\times\hat{a}_n = \left(\frac{J}{2}\right)\hat{a}_x$$

Connection of Concepts for Electrodynamics



3 types of materials

Conductors	Dielectrics	Magnetic
		
Free electrons	Polarized atoms/molecules **Bound electrons**	Magnetic moments **Bound electrons**
		
$\vec{E} = 0$ inside $\rho = 0$ inside $\rho = \rho_s$ only surface charge V is same throughout $\vec{E}_{outside}$ is ⊥ to surface	$\vec{E} \neq 0$ inside but it is reduced $\vec{E}_{tot} = \vec{E}_0 + \vec{E}_{\text{int}}$ $\vec{D} = \epsilon\vec{E}_{\text{tot}} = \vec{P} + \epsilon_0\vec{E}_{tot}$	$\vec{H} \neq 0$ inside but it can be reduced or increased $\vec{B}_{tot} = \vec{B}_0 + \vec{B}_{\text{int}}$ $\vec{B}_{tot} = \mu\vec{H} = \mu_0(\vec{H} + \vec{M})$

Lecture 17 Summary

- Magnetic dipole moment $\mathbf{m}=I\mathbf{A}$
- Magnetization or magnetic dipole moment per unit volume

$$\mathbf{M}=N\mathbf{m}=\chi_m\mathbf{H}_{external}$$

  - Simple linear isotropic magnetic material
    - Strengthens **B**-field strength by $(1+\chi_m)$
    - **H** has same value as free space
  - Magnetization current $\mathbf{J}_M=\nabla x\mathbf{M}$
  - New definition $\mathbf{H}=\mathbf{B}/\mu_0-\mathbf{M}$ and $\mathbf{B}=\mu\mathbf{H}$

- Upcoming schedule: Plane Waves